

★ Easy • #9 • Palindrome Number

▲ Problem Statement • Given an integer "x", return "true" if it is a palindrome, otherwise return "false".

▲ Main Solution • Reverse the number digit by digit & compare number.

▲ Brute Force • Convert the number to a string, reverse it, & compare it with the original string.

▲ Algorithm → 1] If "x" is negative, return "false"

2] Save "x" in "Original".

3] While "x > 0" :-

- Get last digit using "% 10".
- Add it to "reversed".
- Remove the last digit using "/ 10".

4] - Compare "original" & "reversed".

▲ Important Points - "Negative Number are not palindromes"

• " $\%10$ " gets the last digit.

• " $/10$ " removes the last digit

• "long long" helps avoid overflow while reversing.

▲ Complexity - Brute Force - $O(n)$ & $O(n)$

Main Solution - $O(\log_{10} x)$ & $O(1)$